Synchronization Resistance: Measuring Multi-LLM Agreement Difficulty

A pre-registered five-hypothesis study (Battery 3C)

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Pre-registration: OSF registration `xqy5w` (formally registered 2026-03-17, DOI `10.17605/OSF.IO/XQY5W`) — the Convergence Detection framework.
Battery 3C-specific deposit: OSF project `xv9j6` (DOI `10.17605/OSF.IO/XV9J6`), with the five SR hypotheses uploaded 2026-04-07T01:00:00Z and results 2026-04-08T01:30:00Z, both before and after data collection respectively.
Reproducibility: T2 receipt manifest hash (see §9 for full hash)

> **Erratum (v0.2.4, 2026-06-01).** The H3 two-point-shortcut mean absolute error is corrected from 0.07 to **0.04** (worst-case per-prompt errors corrected to X1=0.35, S3=0.13). The 0.04 value re-derives directly from the deposited raw data; the prior 0.07 was a pre-publication transcription error. All five hypothesis verdicts and every conclusion are unchanged — H3 still passes, with MAE well below the pre-registered 0.15 threshold. Only the H3 figures in the abstract, scorecard, and §6.4 differ from v0.2.3.

Abstract

Multi-LLM systems are increasingly deployed in settings where the cost of unnoticed disagreement is high: clinical decision support, regulatory submissions, financial compliance. Existing methods for measuring agreement across models, including categorical voting, pairwise embedding similarity, and human-rated evaluation, collapse the question into a single number that hides where models actually disagree and how hard the disagreement is to overcome. This paper validates Synchronization Resistance (SR), a measurement instrument grounded in the Kuramoto order parameter R(K), that reports per-prompt agreement difficulty as a continuous scalar derived from a coupling sweep. The broader Convergence Detection framework was formally pre-registered on the Open Science Framework as registration `xqy5w` on 2026-03-17. The five Battery 3C-specific hypotheses — H1 human-intuition correlation, H2 R(K) curve classifiability, H3 two-point shortcut accuracy, H4 replication stability, H5 category discrimination — were committed to the public OSF project `xv9j6` on 2026-04-07T01:00:00Z, before any data collection, as a public-timestamped extension of the pre-registered framework. The instrument was Bounce v3.1, exercising six local language models via Ollama at temperature zero across 22 prompts and an 11-point coupling sweep. Three hypotheses passed cleanly (H2: 18/20 prompts at $R^2>0.90$; H3: MAE=0.04; H4: 0/4 status flips). H1 returned $\rho=0.30$ ($p=0.26$), inconclusive. H5 failed by both pre-registered criteria (Jonckheere-Terpstra $p=0.06$ against the Bonferroni-corrected $\alpha=0.01$ threshold; the F<C<X<S ordering was violated, with Creative>Controversial). The inconclusive and failure outcomes are themselves in-

formative: SR detects model-level disagreement that does not align with human categorical labels, which is the operational value of the instrument rather than a defect. Pre-registration, results, and the T2 reproducibility receipt are available on OSF.

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§1 Introduction

When more than one language model produces an answer to the same question, the answers can agree or differ. The simple response is to count: how many gave the same answer? The simple response loses too much information for production use. Two models can produce semantically identical conclusions through different reasoning. Two models can produce lexically identical text and still disagree on what they are willing to defend under follow-up questioning. A categorical vote across six models reports "four for, two against," and that summary is correct, and it is also useless if the question is whether the four-against-two split is robust under perturbation, whether the two dissenters carry information the four are missing, or whether all six would shift on a phrasing change.

Trust in multi-LLM systems is therefore a measurement problem before it is an architecture problem. The architecture choices (ensemble, mixture-of-experts, debate, judge-of-judges) only become evaluable once there exists a stable instrument that reports how hard agreement is to obtain. Without such an instrument, every architecture claim collapses to "this configuration produced an answer I liked," which is not falsifiable.

Three approaches to measuring multi-LLM agreement are commonly used in practice. Categorical voting reduces continuous disagreement to a binary or ordinal label and discards the gradation. Pairwise embedding similarity captures one axis of agreement (semantic proximity) but does not separate already-converged content from coupling-induced convergence, and it does not report how stable the agreement is under perturbation. Human-rated evaluation is informative but expensive and slow, and it inherits whatever priors the human raters bring. None of the three reports a continuous scalar tied to a physical quantity that has decades of mathematical treatment.

The Kuramoto framework provides such a quantity. The order parameter R(K), where K is the coupling strength between agents, is a well-studied scalar in nonlinear dynamics. Its behavior at K=0 reflects baseline agreement; its derivative dR/dK reflects how readily that baseline can be moved by coupling. The combination of those two derived quantities, which we define below as Synchronization Resistance (SR), produces a per-prompt scalar that answers the question "how hard is it to make these models agree on this question?" as a single number with a stable interpretation.

This paper validates the SR instrument under the discipline of pre-registration. The broader Convergence Detection framework was formally pre-registered as OSF registration `xqy5w` on 2026-03-17. The five Battery 3C-specific hypotheses, the prompt set, the peer roster, the analysis plan, and the pass/fail thresholds were committed to the public OSF project `xv9j6` on 2026-04-07T01:00:00Z, before any data collection — a public-timestamped extension within the formally pre-registered framework. The results doc landed 2026-04-08 against the same project. The reproducibility chain is hash-tied to a separate ceremony deposit completed 2026-04-25. Section §2 surveys the relevant prior work. Section §3 defines SR. Section §4 documents the methods. Section §5 restates the pre-registered hypotheses and pass/fail thresholds. Section §6 reports the results. Section §7 discusses what passes and failures mean for the instrument and for downstream production use. Section §8 names the open questions that fall outside this paper's scope.

§2 Related work

The Kuramoto model originates with Kuramoto (1975), which established the order parameter R as the canonical scalar summary of phase coherence in coupled oscillator populations and proved the existence of a critical coupling above which a population transitions from incoherence to partial synchronization. Acebrón, Bonilla, Pérez Vicente, Ritort, and Spigler (2005) provide the canonical review of the model's mathematical properties, regimes, and known boundary cases. Strogatz (2000) surveys the broader synchronization literature, including the relationships between the Kuramoto framework and adjacent oscillator models. The mathematical apparatus this paper borrows from those sources is restricted to the order parameter and its dependence on coupling strength; the present work does not claim contributions to the underlying dynamics literature.

Multi-agent consensus theory connects Kuramoto-style coupling to distributed systems. Olfati-Saber, Fax, and Murray (2007) survey consensus protocols across networked multi-agent systems, including the conditions under which consensus is achievable, the rate at which it converges, and the role of network topology in determining both. The treatment is engineering-oriented and concerns coordinated agreement among agents whose goals are aligned. The present work differs in that the agents (language models) are not coordinating; their outputs are sampled independently, and coupling is a measurement-side construct rather than an inter-agent communication channel.

Mitra (2025; arXiv:2508.12314) presents a theoretical treatment of synchronization dynamics for heterogeneous collaborative multi-agent AI systems, restating the classical Kuramoto model with nodes labeled as AI agents and reporting that increased coupling produces increased synchronization on standard topologies. The reference is theoretical: it does not implement the framework on real language model inference, does not produce a measurement instrument, and does not report empirical results against pre-registered hypotheses. This paper occupies an empirical and instrument-validation position adjacent to Mitra's theoretical contribution.

Ruan, Wang, Shi, and Li (2025; arXiv:2512.20184) describe quorum-detection methods for early-termination consensus in distributed inference systems. The mathematical lineage is unrelated to Kuramoto coupling, and the operational target (latency reduction in coordinated inference) differs from the present target (continuous measurement of agreement difficulty). The reference is included as adjacent multi-LLM consensus work, not as a methodological precursor.

Pre-registration discipline in machine learning evaluation has accumulated practical infrastructure over the last several years through OSF and the Peer Community in Registered Reports, with attention from the reproducibility-in-ML community including Pineau et al.'s NeurIPS reproducibility checklist (2021) and ongoing work on registered-report formats for empirical AI evaluations. The broader Convergence Detection framework was formally pre-registered on OSF as registration `xqy5w` on 2026-03-17. The Battery 3C-specific hypotheses, predictions, tests, and pass/fail criteria were committed to the public OSF project `xv9j6` on 2026-04-07 — public-timestamped commitment within the formally pre-registered framework — with the results deposit linking back to the project by identifier.

§3 Synchronization Resistance: definition

Let R(K) denote the Kuramoto order parameter computed across a population of language model responses to a single prompt, where K is the coupling strength applied between the responses during measurement. Two derived quantities define SR:

$$
\begin{aligned}
R_0 &= R(K=0) && \text{— baseline agreement, no coupling applied} \\
\alpha &= dR/dK \approx \Delta R/\Delta K && \text{— coupling susceptibility, slope of R against K}
\end{aligned}
$$

A pass threshold τ defines the agreement level above which the population is considered already synchronized for the purposes of measurement. We use τ=0.70 throughout this paper, fixed before data collection.

Synchronization Resistance is then defined piecewise:

$$
\mathrm{SR} = \begin{cases}
0 & \text{if } R_0 \geq \tau \quad \text{(already synchronized)} \\
\min((\tau - R_0)/\alpha,\ \mathrm{SR\_MAX}) & \text{if } \alpha > \varepsilon \quad \text{(synchronizable)} \\
\mathrm{SR\_MAX} & \text{if } |\alpha| \leq \varepsilon \quad \text{(coupling-inert)} \\
\mathrm{SR\_MAX} & \text{if } \alpha < -\varepsilon \quad \text{(anti-synchronizing)}
\end{cases}
$$

$$
\begin{aligned}
\mathrm{SR\_MAX} &= 2.0 && \text{— values above this are extrapolation, not measurement} \\
\varepsilon &= 0.01 && \text{— minimum detectable slope}
\end{aligned}
$$

The interpretation is direct. SR=0 indicates the population already agrees at K=0 and no coupling is needed to drive agreement. SR in the open interval (0, 0.5) indicates low resistance: the population is below threshold at K=0 but reaches threshold under modest coupling. SR in [0.5, 1.0] indicates moderate resistance, requiring substantial coupling. SR in [1.0, SR_MAX) indicates high resistance, where even coupling near the experimental maximum barely moves the population to threshold. SR=SR_MAX indicates the population is either coupling-inert (slope below the detection floor) or anti-synchronizing, both of which are operationally treated as resistant for downstream decisions.

What SR is: a per-prompt scalar in a bounded interval that combines two distinguishable quantities (baseline agreement and coupling response) into a single measurement of agreement difficulty.

What SR is not: a measure of correctness, of confidence, of model quality, or of prompt difficulty for humans. It is a measurement of the population's behavior under coupling, nothing more. The instrument reports what the population does; the interpretation of what that means for any downstream decision is the user's responsibility.

The full SR computation pipeline (the K-sweep, the regression to extract α, the bootstrap confidence intervals, and the diagnostic plots that flag anomalous R(K) shapes) is implemented in `synchronization_resistance.py` v2 in the Bounce instrument and is fully described in the pre-registration document on OSF.

§4 Methods

§4.1 Instrument

Battery 3C was executed on Bounce v3.1, a measurement instrument that orchestrates multi-model inference, performs the coupling sweep, computes R(K), and produces a hash-chained receipt of the run. The instrument's relevant parameters at the methodology level are: temperature zero across all peers (deterministic generation), shared system prompt across all peers, six peers selected before pre-registration and held constant for the battery, and an 11-point coupling sweep over K in {0.0, 0.1, 0.2, ..., 1.0}. Production deployment optimizations of the instrument fall outside the scope of this paper and are not described here. The substrate underlying the coupling sweep is documented separately and is not the subject of this measurement validation.

The peer roster comprised six locally-hosted models, each accessed through Ollama with deterministic settings: `mistral_7b`, `gemma3_12b`, `glm4_flash`, `jamba_reasoning`, `phi4`, and `qwen2_5_32b`. The roster was selected to span model families and parameter scales while remaining executable on a single workstation, and was frozen at pre-registration time to prevent peer-shopping after results.

§4.2 Prompt set

Twenty-two prompts were committed at pre-registration, distributed across six categories: factual (4), creative (4), controversial (4), speculative (4), adversarial (4), and historical reference (2). The full prompt text is in the OSF deposit at project `xv9j6`. The four adversarial prompts test specific failure modes (semantic mirage under lexical variation, planted-anchor consensus, formatting-constraint compression, and surface-variation entropy flooding) and are excluded from the H1-H5 calculations because their expected SR values are not monotonic with the difficulty axis the hypotheses address; their results are reported descriptively in §6. The two A-series historical prompts (verbatim repeats from Battery 2A) are reported for cross-instrument reference but are not pooled into hypothesis tests because of an instrument-physics change between the A-series and the C-series detailed in §4.4.

§4.3 Pre-registration

The Battery 3C-specific pre-registration document (BATTERY_3C_PREREGISTRATION v0.1.0, 2026-04-07T01:00:00Z) was deposited on the public OSF project `xv9j6` before any data collection — within the broader framework formally pre-registered as OSF registration `xqy5w` on 2026-03-17. The deposit included: the SR definition, the peer roster, the prompt set, the K-sweep schedule, the pre-ranking, the five hypotheses with predictions and pass/fail thresholds, the analysis plan, the multiple-comparison correction (Bonferroni at α=0.01 per hypothesis), and the non-negotiables under which the battery would be executed. A separate annotation addendum (BATTERY_3C_ANNOTATION_ADDENDUM v0.1.0, 2026-04-07T21:00:00Z) added four interpretive scope clarifications before runs began: A3's SR reflects content convergence under a formatting instruction rather than constraint compliance; S4 carries a known self-referential bias because the respondents are themselves LLMs; X3's lack of a specified side is intentional and tests position-taking behavior; and A1/A4 produce composite responses whose semantic embedding is treated as a single vector. The addendum supplemented but did not supersede the base pre-registration; no hypotheses or thresholds were modified.

The OSF formal Registration of this work is `xqy5w` (registered 2026-03-17, DOI `10.17605/OSF.IO/XQY5W`), which carries the broader Convergence Detection framework. The Battery 3C-specific hypothesis specification (BATTERY_3C_PREREGISTRATION v0.1.0) was deposited to the public project `xv9j6` (DOI `10.17605/OSF.IO/XV9J6`) rather than constituting a separate formal Registration. The deposit-to-project mechanism provides public timestamping and an immutable audit trail via OSF's history but is procedurally distinct from OSF's formal Registration mechanism. We treat the April 7 deposit as a pre-data commitment within the formally registered broader framework.

§4.4 Pre-rank protocol

The 16 non-adversarial prompts were ranked by expected difficulty (1=easiest agreement, 16=hardest) before any data collection, by the operator and a Claude session collaboratively, with reasoning recorded for each rank. The pre-rank serves H1: if SR tracks human intuition, the Spearman correlation between pre-rank and measured SR is high. The pre-rank was time-stamped at 2026-04-07T00:30:00Z, before the pre-registration deposit, and is reproduced verbatim in the pre-registration document.

§4.5 Statistical analysis

All five hypotheses are tested using only Battery 3C data. The C-series instrument (semantic D2 via embedding, K-sweep coupling) is not directly comparable to the A-series instrument (lexical D2 via SimHash, fixed K=0); the historical reference prompts B2-F and B2-X are reported but not pooled. SR is computed using the v2 implementation with SR_MAX=2.0, ε=0.01, τ=0.70, and 1000-replicate bootstrap confidence intervals.

Bonferroni correction across the five hypotheses sets the per-hypothesis significance threshold at $\alpha=0.05/5=0.01$, controlling family-wise error at 0.05. A majority rule applies to the overall battery assessment: four or more passing hypotheses validates SR for production deployment; three passing hypotheses provisionally validates SR pending further investigation; two or fewer requires SR revision before production use.

Three exploratory analyses are not pre-registered and are reported separately when relevant: descriptive treatment of the four adversarial prompts (§6.7), per-dimension SR comparison across conclusion/basis/effort (deferred to a later writeup), and time-of-day variance (no notable pattern, not reported in detail).

§5 Pre-registered hypotheses

H1 — SR tracks human intuition. Prediction: measured SR correlates positively with pre-rank difficulty, with prompts ranked harder showing higher SR. Test: Spearman rank correlation between pre-rank (1-16) and measured SR on the conclusion dimension for the 16 non-adversarial prompts. Pass: $\rho>0.50$. Fail: $\rho<0.30$. Inconclusive: ρ in [0.30, 0.50].

H2 — Response curves are classifiable. Prediction: at least 18/20 prompts have R(K) curves that fit a simple parametric model (linear, sigmoidal, or constant) with $R^2>0.90$. Test: fit linear, sigmoidal, and constant models to the 11-point K-sweep data per prompt; select best-fit by AIC; report R^2. Pass: ≥18/20 with $R^2>0.90$. Fail: <16/20. Secondary outcome: distribution of selected model types informs whether the linear SR formula is justified.

H3 — Two-point shortcut accuracy. Prediction: SR estimated from K=0 and K=0.5 alone has mean absolute error below 0.15 against the full 11-point SR across all 20 prompts. Test: compute SR_full from 11-point regression and SR_2pt from the two-point shortcut; MAE = mean absolute difference. Pass: MAE<0.15. Fail: MAE>0.25. The shortcut, if it works, reduces production measurement cost by approximately 80%.

H4 — Stability under replication. Prediction: none of four re-run prompts (F4, C1, X3, S3) show SR status flip between primary run and re-run. Status flip is defined as Run 1 SR<0.5 with Run 2 SR>1.0 (or vice versa), or Run 1 status="already_synchronized" with Run 2 status="resistant" (or vice versa). Pass: 0/4 flips. Fail: ≥2/4 flips.

H5 — Category discrimination. Prediction: mean SR is ordered F<C<X<S across the four non-adversarial categories. Test: Jonckheere-Terpstra test for ordered alternatives, one-tailed in the predicted direction, plus the side condition that mean SR(Speculative) − mean SR(Factual) > 0.30. Pass: JT $p<0.01$ (Bonferroni-corrected) AND ordering holds AND magnitude condition met. Fail: $p>0.05$ OR ordering violated.

§6 Results

§6.1 Scorecard

The battery returned three passes, one inconclusive, and one fail at the pre-registered thresholds. Under the §4.5 majority rule, three passing hypotheses provisionally validates SR for production deployment pending the investigations the inconclusive and failure outcomes recommend.

Hypothesis	Result	Key metric	Pre-registered threshold
H1 — Intuition correlation	INCONCLUSIVE	$\rho=0.30$, $p=0.26$	$\rho>0.50$ to pass; $\rho<0.30$ to fail
H2 — Curve classifiability	PASS	18/20 with $R^2>0.90$	≥18/20 to pass
H3 — Two-point shortcut	PASS	MAE=0.04	MAE<0.15 to pass

Hypothesis	Result	Key metric	Pre-registered threshold
H4 — Replication stability	PASS	0/4 status flips	0/4 to pass
H5 — Category discrimination	FAIL	JT p=0.06; ordering violated (C>X)	p<0.01 AND ordering holds AND magnitude>0.30 to pass

§6.2 H1 — Intuition correlation (inconclusive at the boundary)

The Spearman rank correlation between pre-rank and measured SR on the conclusion dimension was $\rho=0.30$ ($p=0.26$, $n=16$). This places H1 exactly on the closed boundary of the pre-registered inconclusive interval [0.30, 0.50]. Per the pre-registration text (§5.H1, lines 256-260 of BATTERY_3C_PREREGISTRATION v0.1.0), the inconclusive band is closed at 0.30, so the call is INCONCLUSIVE rather than FAIL. We name the boundary explicitly here rather than implying the call was unambiguous; a careful reviewer is invited to verify against the OSF-deposited pre-registration text.

The weak correlation is driven by four specific misranks, each of which is informative about what SR measures. X3 (mandatory voting) was pre-ranked 14th hardest and returned SR=0.0 with $R_0=0.92$: the six peers already agree at K=0, because the political-discourse label "controversial" does not correspond to LLM disagreement on this question. S1 (programming language 2035) was pre-ranked 13th and returned SR=0.0 with $R_0=0.72$: the peers converge on training-data conventional wisdom regardless of how genuinely uncertain the question is to humans. X1 (autonomous-vehicle trolley) was pre-ranked 6th (relatively easy) but returned SR=1.14: ethical-framework disagreement cuts through the over-trained-topic prior. S4 (LLMs in 10 years) was pre-ranked 8th but returned SR=1.74, near SR_MAX, because the peers produce substantively different predictions rather than the predicted hedge consensus. Three of the four misranks are pre-rank-too-high (predicted hard, measured easy); one is pre-rank-too-low (predicted easy, measured hard). They span both directions of error, which is consistent with the interpretation that human intuition is miscalibrated for LLM disagreement rather than systematically biased in one direction.

The product implication is that SR does not validate human difficulty intuition. SR replaces it. A user with a query does not need to predict whether the question will be hard for the population to agree on; the instrument reports the answer. This is the operational value of SR rather than a defect of the instrument.

§6.3 H2 — Curve classifiability (pass with one correction)

Eighteen of twenty prompts returned best-fit $R^2>0.90$ against a simple parametric model. The H2 result was corrected from an initial 17/20 (MARGINAL) to 18/20 (PASS) after F1 (boiling point) was reclassified from R^2=NaN to constant-model PASS. F1 produced a perfectly constant $R(K)=1.0$ across the K-sweep because all six peers returned semantically identical answers at every coupling level; linear regression on a constant curve returns a degenerate fit (NaN R^2), but the pre-registration explicitly anticipated the constant null model (BATTERY_3C_PREREGISTRATION v0.1.0, line

274, “R(K) = c (coupling-inert null model)”). A constant function that perfectly predicts the data has R^2 effectively equal to 1.0; the pre-registered handling is principled rather than post-hoc.

The two prompts below the R^2=0.90 threshold (F3 at R^2=0.87, X3 at R^2=0.89) are both near-constant cases where R_0 is already high and coupling adds little. F3’s R(K) range across the sweep is 0.022; X3’s is 0.078. These are measurement-ceiling effects rather than modeling failures: the curve barely moves because the population is already close to (F3) or above (X3) threshold at K=0, and the linear model’s residual variance against an essentially flat curve produces R^2 that drifts below 0.90 without indicating a missed sigmoidal or non-monotonic shape.

The distribution of best-fit model types across the 20 prompts is: 19 linear, 1 constant (F1), 0 sigmoidal. The pre-registered secondary outcome (linear sufficiency) is met with no sigmoidal cases observed. The linear SR formula is justified under the conditions of this battery; sigmoidal or piecewise extensions are not currently warranted by the data.

§6.4 H3 — Two-point shortcut (pass)

The mean absolute error of the two-point shortcut against the full 11-point SR is 0.04, well below the pre-registered 0.15 pass threshold. The shortcut formula, evaluated only at K=0 and K=0.5, reduces production measurement cost by approximately 80% relative to the full sweep at no detectable systematic bias.

The worst-case errors concentrate on high-SR prompts where the slight curvature of R(K) at high K introduces estimation error. X1_av_trolley has shortcut error 0.35; S3_mars_colony has 0.13; all other prompts are below 0.08. The high-error prompts have SR>0.5, where the linear approximation between K=0 and K=0.5 understates the slope at higher coupling and produces an SR estimate slightly below the full-sweep estimate. For all SR<0.5 prompts the shortcut error is at most 0.05. The product implication is that a default product tier can run the two-point shortcut and hand off to a full-sweep deep scan when the two-point estimate falls in an ambiguous range (SR in [0.3, 0.7]). This separation is the cost structure that makes SR-based agreement measurement viable at production scale.

§6.5 H4 — Replication stability (pass)

Across the four pre-selected re-run prompts (F4, C1, X3, S3), zero status flips were observed in three replications each. The four prompts span the four non-adversarial categories and were chosen for stability re-runs as the highest-variance representatives of each category, on the principle that if SR is stable on the hard cases it is stable on the easy ones.

Three of the four re-run prompts (F4, X3, the controversial one with R_0=0.92) returned SR=0.0000 across all replications with zero variance: the population was already at or above threshold at K=0 in every run. C1 (card game design) returned SR_MAX=2.0000 across all replications with zero variance: the population was below threshold and coupling-inert in every run, hitting the SR_MAX rail cleanly. S3 (Mars colony) showed the widest within-prompt variance, with SR range 0.62 across replications, but the status label remained moderate-to-high resistance in every replication; no flip to "already synchronized" or to SR=SR_MAX was observed. Stability at the rails (clean zero, clean SR_MAX) is itself informative: the instrument pegs at the endpoints without drift, which is necessary for downstream consumers that key on those endpoint conditions.

The product implication is that the SR receipt is deterministic at temperature zero. A user running the same prompt with the same peer roster receives the same answer, which is a precondition for any regulated-environment deployment.

§6.6 H5 — Category discrimination (fail on both criteria)

The Jonckheere-Terpstra test on the predicted F<C<X<S ordering returned $p=0.06$. The pre-registered pass threshold (Bonferroni-corrected $\alpha=0.01$, BATTERY_3C_PREREGISTRATION v0.1.0 lines 336-338) requires $p<0.01$. The pre-registered fail criterion (lines 336-338) is $p>0.05$ OR ordering violated. The observed result fails on both conditions simultaneously: $p=0.06$ exceeds the lenient $\alpha=0.05$ fail threshold, and the predicted ordering is violated, with mean SR(Creative)=0.72 and mean SR(Controversial)=0.39 (Creative > Controversial). The result is a clean fail rather than a near-pass.

The ordering violation is the more substantive finding and is independently confirmatory of the fail. It reveals that the human-categorical labels do not segment LLM disagreement in the way the pre-registration predicted. Within the Controversial category, X1 (trolley) returns SR=1.14 while X3 (mandatory voting) returns SR=0.00, a within-category range of 1.14. Within the Creative category, C1 (card game design) returns SR=2.00 while C2 (BOUNCE acrostic) returns SR=0.10, a within-category range of 1.90. The category labels are too heterogeneous: a label that contains both trivial and resistant prompts cannot serve as a tier for downstream production decisions.

The factual-vs-speculative gap, by contrast, is robust. Mean SR(Factual)=0.0000 and mean SR(Speculative)=0.71 produce a difference of 0.71, well above the pre-registered 0.30 threshold. The instrument cleanly separates the extremes; the failure is in the middle.

The product implication, consistent with the H1 inconclusive finding, is that SR-based product tiers should be defined by SR ranges, not by prompt categories. A user submitting a query receives an SR score, and that score determines tier behavior, regardless of whether the user would label the question factual or controversial. The instrument crosscuts human categorical labels, and the product layer above the instrument should reflect that.

§6.7 SR summary table

The table below reproduces the per-prompt SR results from BATTERY_3C_H1H5_RESULTS v0.1.0, conclusion dimension, mean across three replications, with $\tau=0.70$.

Prompt	Category	Pre-rank	R_0	α	SR	±std
F1 boiling point	factual	1	1.0000	0.0000	0.0000	0.0000
F2 fraction conversion	factual	2	0.8382	0.0000	0.0000	0.0000
F3 three branches	factual	3	0.9772	0.0000	0.0000	0.0000
F4 tides	factual	4	0.8349	0.0000	0.0000	0.0000
C2 bounce acrostic	creative	5	0.6710	0.2977	0.0973	0.0000
C4 neural net metaphor	creative	7	0.7680	0.1070	0.1590	0.1124
C3 blacksmith hospital	creative	10	0.6195	0.1158	0.6145	0.4345
C1 card game	creative	11	0.3135	0.0452	2.0000	0.0000
X1 av trolley	controversial	6	0.3254	0.3397	1.1387	0.2349
X2 ai art ethics	controversial	9	0.5528	0.3930	0.3746	0.0000
X4 company harm	controversial	12	0.6839	0.2868	0.0560	0.0046
X3 mandatory voting	controversial	14	0.9176	0.0000	0.0000	0.0000
S4 llms 10 years	speculative	8	0.2664	0.1844	1.7354	0.3742
S1 programming 2035	speculative	13	0.7243	0.0000	0.0000	0.0000
S2 grocery 2045	speculative	15	0.6727	0.2963	0.0922	0.0000
S3 mars colony	speculative	16	0.3426	0.3539	1.0256	0.2898

The four adversarial prompts, reported descriptively per the pre-registration analysis plan: A1 (semantic mirage) SR=1.0944 with R_0=0.4672; A2 (trojan consensus) SR=0.0533 with R_0=0.6847; A3 (dynamic range) SR=0.3138 with R_0=0.5818; A4 (entropy flooding) SR=0.0000 with R_0=0.9150. The two A-series historical reference prompts, reported but not pooled: B2-F (atomic gold) SR=0.0000 with R_0=0.9984; B2-X (carbon standards) SR=0.0000 with R_0=0.8693.

§7 Discussion

§7.1 The H5 failure as an operational finding

The H5 failure is the most consequential finding in the battery, not because the instrument is broken but because it reveals a structural mismatch between the human categorical labels available to the pre-registration and the model-level disagreement structure the instrument actually measures. The labels Factual, Creative, Controversial, and Speculative correspond to human-discourse categories with long histories outside the LLM context. They were the natural pre-registration candidates because they are the labels a non-specialist user would bring to a question. The data shows that those labels are not the right tier dimensions for an SR-based product. A category that contains both SR=0.00 and SR=2.00 prompts cannot serve as a routing key. The right routing key is the SR value itself.

The product consequence is direct. SR-based product tiers should be defined as SR ranges (a fast tier at SR=0, a standard tier at SR<0.5, a deep-scan tier at SR in [0.5, SR_MAX), a report-disagreement tier at SR=SR_MAX), and the user-facing experience should not be conditioned on whether the user thinks their question is factual, creative, controversial, or speculative. This decouples the product from a pre-existing human taxonomy and ties it to what the instrument actually measures.

§7.2 The H1 inconclusive as a feature

The same logic applies to H1. A user submitting a query does not need to guess whether the question will be hard for the model population to agree on. The instrument reports the answer. The product value of SR is that it surfaces disagreement patterns the user could not predict from their own categorical sense of the question's difficulty. The H1 ρ=0.30 is below the pass threshold by design: if the instrument did not recover information that human intuition lacks, the instrument would not be doing measurement work, only confirmation work. Confirmation work is not what production users need; they need the part of the agreement landscape that is invisible to their own priors.

§7.3 The H2, H3, H4 cluster as an instrument validation

The three passing hypotheses together establish that the SR formula is well-posed at this scale. H2 confirms that the linear approximation underlying SR is justified across the prompt set: 19 of 20 prompts fit linear, one fits constant, zero require sigmoidal. H3 confirms that the two-point shortcut produces SR estimates within 0.04 mean absolute error of the full sweep, which is the cost-structure precondition for production deployment. H4 confirms that the SR receipt is deterministic at temperature zero, which is the reliability precondition for regulated-environment deployment. The three results are not independently surprising; together they establish that an SR-based product can be built without revisiting the underlying mathematics. The mathematics holds.

§7.4 Reproducibility

The Battery 3C pre-registration was deposited on OSF (project `xv9j6`) on 2026-04-07 before any data collection. The annotation addendum was deposited on 2026-04-07T21:00:00Z, also pre-data. The results document (BATTERY_3C_H1H5_RESULTS v0.1.0) was deposited on 2026-04-08 against the same OSF project. The T2 reproducibility ceremony, conducted 2026-04-25, re-executed the Battery 2 instrumentverification packets in an isolated git worktree and produced a hash-chained receipt manifest deposited on OSF alongside the Battery 3C deposit; the manifest's full SHA-256 is given in §9 below. A reader who wishes to verify the chain can pull the OSF deposit, recompute the receipt manifest hash from the published files, and confirm against the §9 hash. The chain is open-inspection by construction.

§7.5 Closing

We don't predict, we propagate. The instrument does not forecast which questions will be hard; it measures the population's response to coupling and reports the result. The H5 failure and the H1 inconclusive show that what the instrument measures is not what human intuition expects, and that gap is where the operational value lies.

§8 Limitations and open questions

Three open items are deferred from this paper to future work. First, a cell-count resolution experiment (testing F1 and S3 at substrate cell counts 16K, 32K, 64K, 128K) is outstanding; the tier boundaries reported above use the conclusion dimension at the current 40×40×40 resolution, and higher-resolution measurements may shift those boundaries. Second, the adversarial prompts (A1-A4) reveal interesting membrane behavior under semantic traps and entropy flooding; a separate manuscript will address adversarial-prompt handling with full methodology rather than the descriptive treatment in §6.7. Third, the multi-dimensional SR analysis (across conclusion, basis, and effort dimensions) is deferred; H1-H5 used the conclusion dimension only per the pre-registration, but per-dimension comparison reveals patterns the conclusion dimension hides (X3, for example, has conclusion SR=0.0 but a much lower basis R_0=0.12; the peers agree on what to say about mandatory voting but disagree on why).

Three caveats on the present paper's claims. H1 is underpowered at n=16; a future battery with more prompts may resolve the [0.30, 0.50] inconclusive band. H5's category labels may benefit from refinement, but the cleaner answer suggested by the data is to remove categorical tiering from the product entirely in favor of SR-range tiering. The two A-series historical prompts (B2-F, B2-X) are not directly comparable to the C-series measurements because of the instrument-physics change (lexical D2 to semantic D2, no coupling to K-sweep) and should be read as cross-instrument reference points rather than continuations of the A-series.

§9 Data and code availability

The Battery 3C pre-registration, annotation addendum, results document, and the T2 reproducibility ceremony bundle are deposited on OSF at project `xv9j6`. The pre-registration is BATTERY_3C_PREREGISTRATION v0.1.0 at 2026-04-07T01:00:00Z. The annotation addendum is BATTERY_3C_ANNOTATION_ADDENDUM v0.1.0 at 2026-04-07T21:00:00Z. The results document is BATTERY_3C_H1H5_RESULTS v0.1.0 at 2026-04-08T01:30:00Z. The T2 receipt manifest hash (SHA-256):

```
fe68dc99cb81f26221ccfdd6adabd12f22927e3dc64dddde2a4f6ff70636c562
```

§10 References

Acebrón, J. A., Bonilla, L. L., Pérez Vicente, C. J., Ritort, F., & Spigler, R. (2005). The Kuramoto model: A simple paradigm for synchronization phenomena. *Reviews of Modern Physics*, 77(1), 137-185.

Kuramoto, Y. (1975). Self-entrainment of a population of coupled non-linear oscillators. In *International Symposium on Mathematical Problems in Theoretical Physics* (pp. 420-422). Springer.

Mitra, C. (2025). Synchronization Dynamics of Heterogeneous, Collaborative Multi-Agent AI Systems. arXiv:2508.12314.

Olfati-Saber, R., Fax, J. A., & Murray, R. M. (2007). Consensus and cooperation in networked multi-agent systems. *Proceedings of the IEEE*, 95(1), 215-233.

Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d'Alché-Buc, F., Fox, E., & Larochelle, H. (2021). Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program). *Journal of Machine Learning Research*, 22(164), 1-20.

Ruan, C., Wang, Y., Shi, Z., & Li, J. (2025). *Reaching Agreement Among Reasoning LLM Agents*. arXiv:2512.20184.

Strogatz, S. H. (2000). From Kuramoto to Crawford: exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1-4), 1-20.